Status	Finished
Started	Tuesday, 20 October 2026, 10:03 PM
Completed	Tuesday, 20 October 2026, 10:06 PM
Duration	2 mins 43 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes the data unsorted
c. It makes it easier to select items based on their relationship to the rest of the items
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Linear Search
d. Binary Search

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It avoids the need for recursion
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Selection sort
c. Heap sort
d. Merge sort

The correct answer is:
Merge sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Bubble Sort
- c. Quick Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **9**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Insertion
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **10**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. order()
- c. sorted()
- d. sort()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It divides the list into two halves, sorts each half, and then merges them
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is sorted independently
c. Each part is searched for a specific element
d. The parts are ignored

The correct answer is: Each part is sorted independently